Algebra 1
Unit 13
Lesson 9 Homework

Name_Answer Key__________
Date______________________
Period_____________________

To make a 15% acid solution, a chemist mixes 3 parts of 4% acid solution with a 20% acid solution. The expression $3(0.04x) + 0.20y$ represents the mixture, where x is the number of ounces of 4% solution and y is the number of ounces of the 20% solution.

Explain what each quantity means.

1. $(0.04x)$

   Amount of acid solution per part of 4% solution.

2. $3(0.04x)$

   Amount of acid solution mixed into mixture of 4% solution.

3. $0.20y$

   Amount of acid solution per part of 20% solution.

Since 2015, the number of independently contracted food delivery drivers can be modeled by the equation $y = 1{,}250{,}000(1.073)^x$.

4. What does the x variable represent?

   Number of years since 2015.

5. What does the y variable represent?

   Total amount of contracted drivers an "x" number of years after 2015.

6. What does the quantity $1{,}250{,}000$ represent?

   Initial number of drivers in 2015.

7. What is the constant percent rate of change?

   107.3%

The equation $y = -2(x-4)^2 + 45$ models the height, y, in feet, of a ball, x seconds after it is thrown into the air.

Explain what each quantity represents.

8. $x - 4$

   the number of seconds away the ball's height is from the maximum

9. 45

   maximum height of the ball

Jovi is buying cereal for $\$3.25$ per box. He has a coupon for two free cereals. He pays $\$4.75$ for a gallon of milk. The expression $3.25(x-2) + 4.75$ can be used to represent the overall cost.

10. What does the quantity $3.25(x-2)$ represent?

    Cost of cereal boxes purchased.